

AUTOMATED ANNOTATION OF FERRET BURROWING BEHAVIOR USING MACHINE LEARNING

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Introduction

- Behavioral markers are essential for the development of preclinical models
- Burrowing in ferrets is being proposed as a promising naturalistic behavioral marker
- Manual annotation is time-intensive, subjective, and prone to observer drift
- A new pipeline is proposed for the automation of ferret burrowing annotation, allowing for the standardization, identification, and quantification of burrowing in a time-efficient manner



Figure 1: Burrowing setup for in-lab data collection

Methods – Video Collection

- Burrowing area (rice bin) surrounded by playpen
- Camera recording straight down centered on top of the burrowing area
- Recordings were done with a 2k resolution at 60 fps
- 10-minute videos
- 140 videos collected from 16 females and 7 males

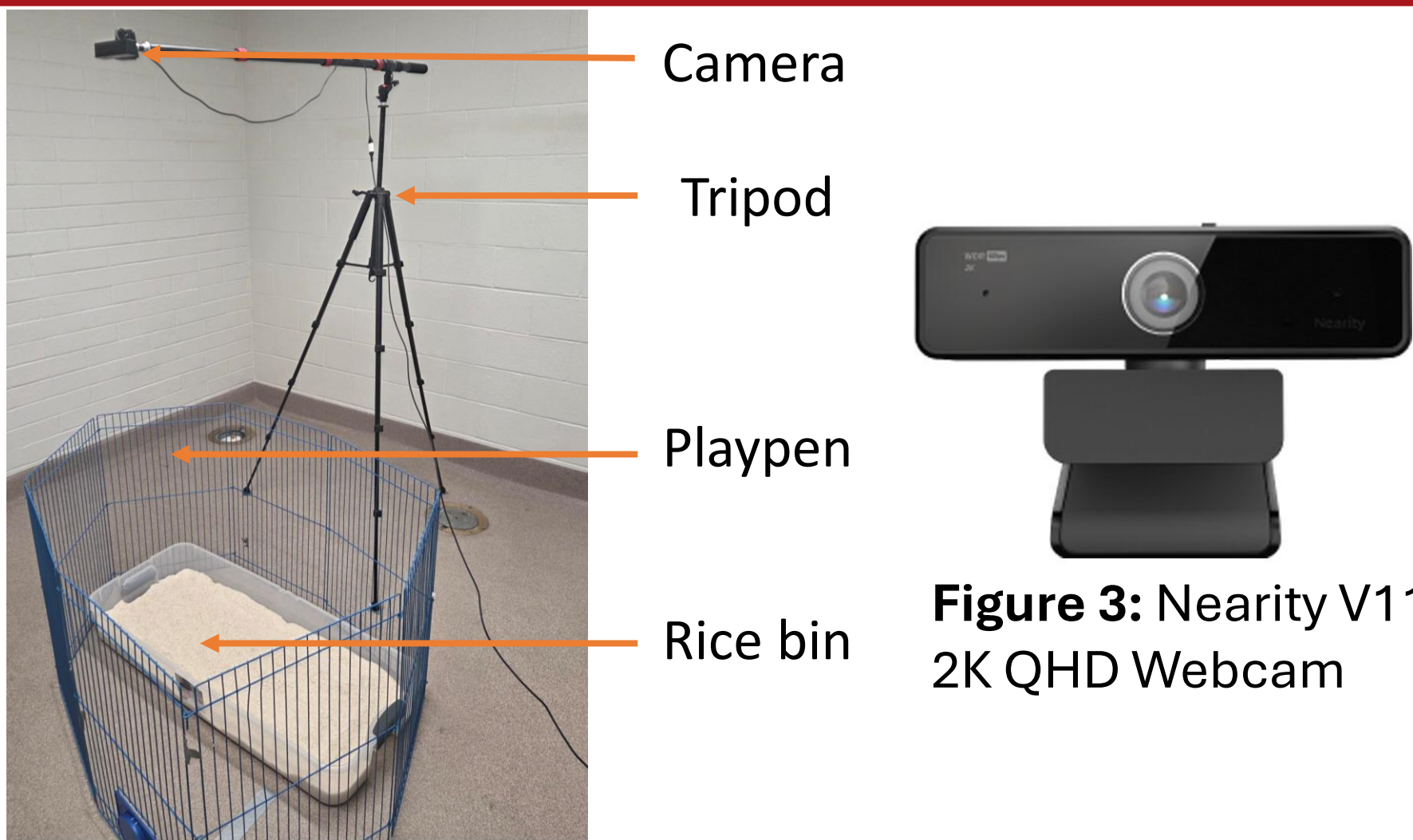
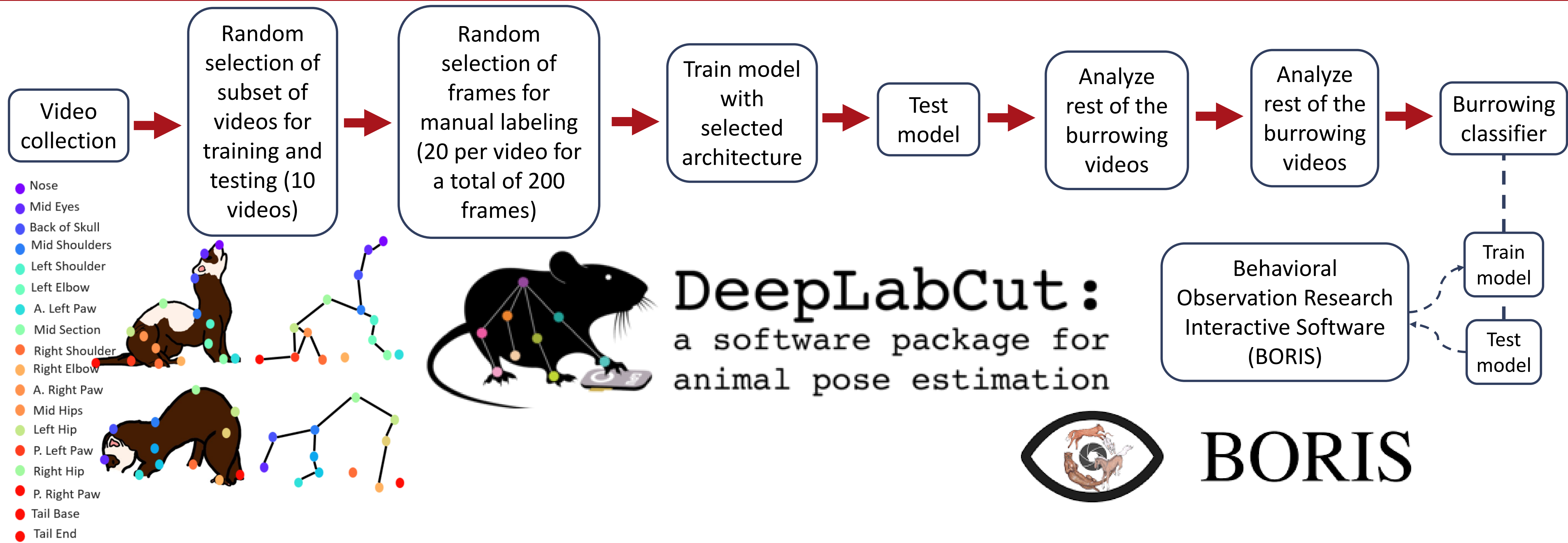


Figure 2: Setup for burrowing video collection in-lab

Methods – Automated Notation Pipeline



Architectures

Architecture	Network Depth	Feature Strategy	Spatial Resolution	Intended Use
ResNet-50	Moderate	Single-scale	Standard	Baseline pose estimation
ResNet-101	Deep	Single-scale	Standard	Complex feature learning
dlcrnet_stride16_ms5	Model-specific	Multi-scale	High	Fine-grained animal pose estimation

Table 1: Three different network architectures were tested for the training of the DeepLabCut model. ResNet-50 is recommended for simple tracking, while ResNet-101 is meant for more complex tracking. Dlcnet_stride16_ms5 builds up from this model but using a multi-scale strategy for higher accuracy.

Ethogram

Behavior type	Behavior code	Description
Digging	DG	Using the front paws to remove the rice while keeping the head up
Burrowing	BG	Inserting the head into the rice and using the front paws to remove the rice while moving forward. Scratching and biting of the inner side of the rice container.
Sniffing	SG	Walking around the inside of the container while keeping the nose close to the rice surface.
Exploratory	EO	Walking, sniffing, and biting and scratching the container while being outside of the enrichment area.
Scratching on the rice	SR	Laying down on the back while moving back and forth.
Scratching/licking	SL	Using the paws, teeth and/or tongue to relief itching while curling the body.
Climbing	C	Climbing over the container walls, using them as support to rest.
Stationary	ST	No movement

Table 2: The ethogram defines the behaviors observed in the videos. These definitions guide the manual annotation in BORIS

Preliminary Results

Architecture	Root Mean Square Error (pixels)
ResNet-50	125.72
ResNet-101	28.37
dlcrnet_stride16_ms5	18.76

Table 3: The table show the test results of the DeepLabCut model when trained with each architecture. The error in pixels indicates how far off was the automated annotation from the manually labeled points

Conclusions and Future Directions

Conclusions:

- Preliminary results showed that the model with the dlcnet_stride16_ms5 architecture has a higher accuracy and will be used for the development of the pipeline

Future directions:

- Analyze all the videos with the DLC model
- Train and test the burrowing classifier
- Analyze the burrowing data

References and Website



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